

EDUCATION

- Arizona State University** Arizona, USA
• *Ph.D. in Computer Science;*
Interest: Reinforcement Learning, LLMs, Sim2Real Transfer, Policy Evaluation
Advisor: Dr. Hua Wei.
August 2023 - May 2026
- Arizona State University** Arizona, USA
• *M.S. in Computer Science;*
Interest: Reinforcement Learning, LLMs
Advisor: Dr. Hua Wei.
August 2023 - December 2025
- New Jersey Institute of Technology** New Jersey, USA
• *Research Assistant of Informatics;*
Interest: Data Mining, Reinforcement Learning.
July 2022 - May 2023

RESEARCH ACHIEVEMENT

Received **KDD'26 Oral and AAAI'24 Oral(<5%)**, **SDM'25 Best Poster Award**, **Google Ph.D. Fellowship Nomination**.

SELECTED PUBLICATIONS, TUTORIALS

- [KDD'26 **Oral (first author)**] **ShadeBench: A Benchmark Dataset for Building Shade Simulation in Sustainable Society**: Proposed the benchmark for urban shade estimation, understand the pros and cons of GenAI for social good.
- [ICML'26] **Position: Uncertainty Quantification in LLMs is Just Unsupervised Clustering**: Discussed the key challenge of the LLM uncertainty quantification in the domain, arguing for a more feasible and comprehensive solution.
- [ICLR'26 (first author)] **Latent Adaptation of Foundation Policies for Sim-to-Real Transfer**: Developed an efficient method for test-time adaptation of foundation policies under domain shifts (in Hilbert Space).
- [AAAI'26 (first author)] **Measuring What Matters: Scenario-Driven Evaluation for Trajectory Predictors in Autonomous Driving**: An adaptive evaluation framework for predictors of autonomous vehicles by balancing the diversity and accuracy in various scenarios, provides a new perspective to decouple and analyze components from complex systems.
- [COLM'25 (first author)] **Understanding the Uncertainty of LLM Explanations: A Perspective Based on Reasoning Topology**: Propose Topo-UQ, by designing a structural elicitation strategy, it decomposes the explanation into the knowledge and reasoning dimensions and allows to quantify reasoning uncertainty and assess knowledge redundancy.
- [KDD'25-a (first author)] **FLanS: A Foundation Model for Free-Form Language-based Segmentation in Medical Images**: A novel medical image segmentation model that handles various free-form text prompts.
- [KDD'25-b **Tutorial**] **Uncertainty Quantification and Confidence Calibration in Large Language Models - A Survey**: A new taxonomy that categorizes uncertainty quantification methods based on computational efficiency and uncertainty dimensions, and assesses their real-world applicability, and open challenges.
- [ICML'25] **Flexible Heteroscedastic Count Regression with Deep Double Poisson Networks**: Propose the Deep Double Poisson Network (DDPN) to produce discrete predictive distributions of arbitrary flexibility, with verified performance on count regression datasets including tabular, image, point cloud, and text data.
- [IJCAI'25-a (first author)] **DeepShade: Enable Shade Simulation by Text-conditioned Image Generation**: Propose a diffusion-based generative model designed to learn and synthesize shade variations over time. It is beneficial for the shade routing systems and public health - AI for social good.
- [IJCAI'25-b (first author)] **GE-Chat: A Graph Enhanced RAG Framework for Evidential Response Generation of LLMs**: Propose GE-Chat, a knowledge Graph-enhanced retrieval-augmented generation framework designed to deliver Evidence-based responses.
- [RLC'25] **Joint-Local Grounded Action Transformation for Sim-to-Real Transfer in Multi-Agent Traffic Control**: Introduced JL-GAT, an application of Grounded Action Transformation to MARL-based TSC that balances scalability with enhanced grounding capability by incorporating information from neighboring agents.
- [SDM'25-a **Best Poster Award (first author)**] **CoMAL: Collaborative Multi-Agent Large Language Models for Mixed-Autonomy Traffic**: An effective LLM-based multi-agent framework for mix-autonomy traffic challenges.
- [SDM'25-b] **Privacy-preserving Fine-tuning of Large Language Models through Flatness**: Address privacy issue in LLM training and fine-tuning process.
- [AAAI'24-a **Oral (first author)**] **Prompt to Transfer: Sim-to-Real Transfer for Traffic Signal Control with Prompt Learning**: Mitigate the sim2real transfer performance impairment in Reinforcement Learning by leveraging LLMs' inference ability.
- [AAAI'24-b (first author)] **Probabilistic Offline Policy Ranking with Approximate Bayesian Computation**: Offline policy evaluation for Reinforcement Learning methods.
- [NeurIPS'24] **RegExplainer: Generating Explanations for Graph Neural Networks in Regression Tasks**: An explanation method to interpret the graph regression models based on graph information bottleneck theory.

- [CIKM'24 ([first author](#))] **Shaded Route Planning Using Active Segmentation and Identification of Satellite Images**: A sustainable routing strategy for residents under heatwaves using foundation models and satellite images.
- [ECML-PKDD'24 ([first author](#))] **CityFlowER: An Efficient and Realistic Traffic Simulator with Embedded Machine Learning Models**: Improve the realism in microscopic level traffic simulation while keeping high efficiency.
- [IJMLC'23 ([first author](#))] **Open-TI: Open Traffic Intelligence with Augmented Language Model**: An open-sourced traffic intelligence using Augmented Language Model.
- [Machine Learning - 2023] **LibSignal: An Open Library for Traffic Signal Control**: Provide cross-simulator training and testing bed for Classic and SOTA algorithms. Also accepted by the NeurIPS 2022 workshop.
- [IEEE-CDC'23 ([first author](#))] **Uncertainty-aware Grounded Action Transformation towards Sim-to-Real Transfer for Traffic Signal Control**: Uncertainty quantification to the action grounding process in sim2real problems.

RESEARCH DIRECTION AND PROJECT

- **LLM assistant for domain intelligence** [IJMLC24, SDM'25-a] and **trustworthiness** [SDM'25-b, IJCAI'25-b, COLM'25]: Identify the feasibility of LLM assistance for general traffic research tasks and identify its uncertainty for better decision-making.
- **Mitigate sim2real problem in RL**: UGAT[CDC'23], **promptGAT**[AAAI'24-a], and **JL-GAT**[RLC'25]: Design methods that help to mitigate performance impairment of simulator learned policy when deployed to real-world scenarios.
- **For a more realistic simulation: synthetic trajectory generation** [ERA'22], **generative AI for shade simulation** [IJCAI'25-a], and **more realistic simulator** [ECML-PKDD'24]: Learn motion features from human moving trajectories and trajectory generation/prediction, and provide a way to embed ML models into the simulator efficiently.
- **Solve the realistic traffic signal control problems in roundabouts - Sedona, AZ, USA**: Funded by Arizona Department of Transportation (ADOT) to solve the congestion problem in Sedona. Apply Genetic Algorithm to optimize the OD-demand matrix (Origin-Destination).
- **An Open-sourced library for Traffic Signal Control**: The development of a cross-simulator comparison platform of reinforcement learning models in traffic signal control tasks.
- **RL and applications: NeurIPS-2022 CityLearn Challenge, Drug Discovery and Design**: Leverage RL/IL algorithms to optimize the building's energy management policy, and provide efficient exploration towards synthesizable chemical structure.

HONORS AND AWARDS

- **2026 Outstanding Graduating CS PhD Student** at Arizona State University - April, 2026.
- **2025 Google Fellowship Nomination** at Arizona State University - June, 2025.
- **Best Poster Award** at Doctoral Forum in SDM 2025 - May, 2025.
- **Top30 Nationwide CS/OR Ph.D. Students** in AI-SCORE summer school **University of Maryland** - May, 2024.
- **Arizona State University Graduate Fellowship (UGF)** - Spring, 2024 and 2025.
- Third prize of the NJIT-DANA KNOX Student Research Showcase (Graduate group) - April, 2023.
- Outstanding student & outstanding graduate thesis - June, 2022.

EXPERIENCE

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| • Invited Researcher at Meta Ph.D. Forum
<i>Invited to connect with Meta leadership and scientists for long-term cooperation.</i> | Meta Headquarters, USA: 2025 Fall |
| • Research Scientist Intern
<i>Spatial Temporal Planning with LLMs in Real-world Routing Systems.</i> | Microsoft, USA: 2025 Summer |
| • Research Intern
<i>Robust Autonomous Vehicles Trajectory Prediction Module</i> | Honda Research, USA: 2025 Spring |
| • AI Scientist Intern
<i>Large Foundation Models for Language Guided Segmentation</i> | GE-Healthcare, USA: 2024 Summer |

RESEARCH COMMUNITY AND SERVICE

- Peer Reviewer of WACV 2025, ACL 2025 Demo, EMNLP 2025, KDD 2025, ICLR 2025.
- Program Committee Member of WWW 2024 and ADMA 2024.
- Peer Reviewer of PAKDD 2024, ICPR 2024, CASE 2023, NeurIPS 2022.
- Reviewer of IEEE Transactions on Intelligent Transportation Systems. 2023.
- Reviewer of IEEE Transactions on Computational Social Systems. 2023.
- Host of the Tutorial in IEEE-ITSC - Bilbao, Spain, 2023.
- SigmaXi Scientific Research Honor Society Member.
- IEEE Regular Membership of Yong Professionals.
- IEEE Regular Membership of Robotics and Automation Society.
- Association for the Advancement of Artificial Intelligence (AAAI) Membership.